

Practice Test IV (Version 1)

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Precalculus).
- II) Show all work.
- III) Draw a rectangle around your final answer for each problem.
- IV) A graphing calculator is required.

① Relations are written below in table form. Some of the tables could represent functions that are one-to-one, some of them could represent functions that are not one-to-one, and some cannot represent functions. For each table, write "1-to-1 function," "function," or "not a function."

i)

x	4	-6	2	0	1
y	9	3	4	1	1

Function

ii)

x	3	4	5	6	7
y	9	10	11	12	13

1-to-1 Function

iii)

x	8	5	3	1	0
y	-8	-9	-10	-11	0

1-to-1 Function

iv)

x	-18	20	4	24	-18
y	0	6	3	9	4

Not a Function

② Each table below represents a function "f." Write the values of the inverse function "f⁻¹." If "f" is not a one-to-one function, then you must omit values of "f⁻¹" so that it qualifies as a function. To ensure that the functions are perfectly matched inverses, values of "f" may need to be omitted also. You may choose which values to omit. Highlight or circle the omitted values.

i)

x	13	10	-9	8	-7
f(x)	11	12	0	14	12

x	11	12	0	14	12
f ⁻¹ (x)	13	10	-9	8	-7

ii)

x	0	4	10	19	-1
f(x)	10	6	0	-6	-1

x	10	6	0	-6	-1
f ⁻¹ (x)	0	4	10	19	-1

iii) If "f" and "g" are inverse functions, find f(2) if g(-3) = 2.

$$(-3, 2) \rightarrow (2, -3) \quad \boxed{f(2) = -3}$$

iv) If "f" and "f⁻¹" are inverse functions, find f(-7) if f⁻¹(5) = -7.

$$(5, -7) \rightarrow (-7, 5) \quad \boxed{f(-7) = 5}$$

③ Find the mode of the data set below.

80, 79, 81, 83, 81, 81

81

④ A plane flew 412 miles per hour for 5 hours. It then increased its speed to 460 miles per hour and flew for 4 hours. What was the average speed of the plane during the entire distance?

$$\text{Average speed} = \frac{\text{total distance traveled}}{\text{total time passed}} = \frac{5(412) + 4(460)}{5 + 4} = \boxed{433\frac{1}{3} \text{ mi/h}}$$

⑤ For the function below, find an algebraic expression representing the difference quotient. Use "h" for "change in x." It is not necessary to simplify the expression.

$$f(x) = x^2 + \sin x - 1$$

$$DQ = \frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 + \sin(x+h) - 1 - (x^2 + \sin x - 1)}{h}$$

For problem #5 6-7, Find the equation of the inverse function. Restrictions must be used on the original function and/or inverse function so that they are perfectly matched and no aberrations occur. To understand this, graph the function and note that it fails the Horizontal Line Test, which means that it is not one-to-one. Draw the inverse graph and note that it fails the Vertical Line Test. Highlight the parts of the inverse graph and the original graph that you will omit and use inequality notation next to the equations to indicate restrictions.

⑥ $f(x) = x^2 + 3, \quad x \geq 0$

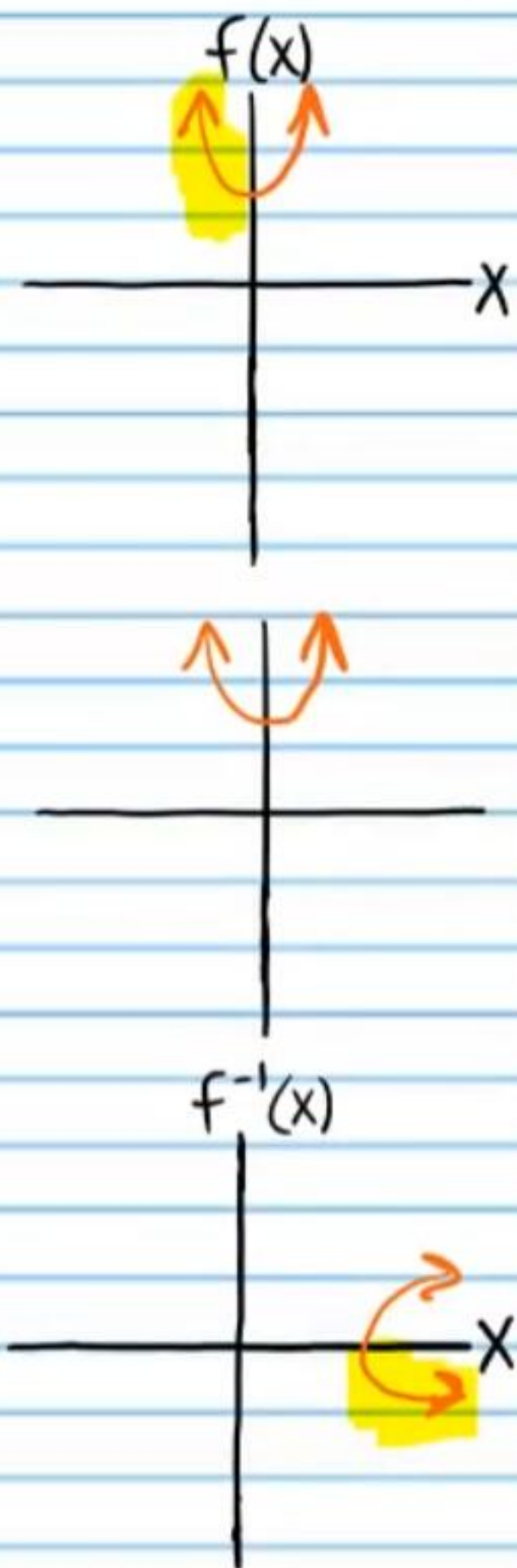
I) $y = x^2 + 3$

II) $x = y^2 + 3$

III) $x - 3 = y^2$

$\pm\sqrt{x-3} = y$

$f^{-1}(x) = \sqrt{x-3}$



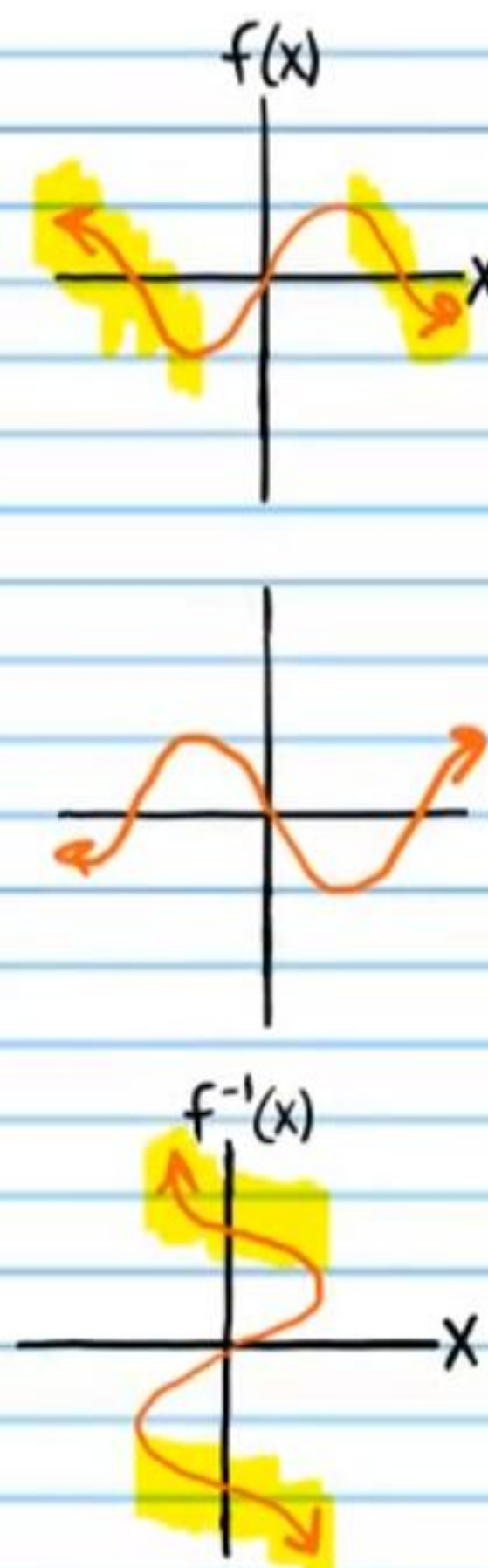
⑦ $f(x) = \sin x, \quad -90^\circ \leq x \leq 90^\circ$

I) $y = \sin x$

II) $x = \sin y$

III) $y = \sin^{-1} x$

IV) $f^{-1}(x) = \sin^{-1} x, \quad -90^\circ \leq y \leq 90^\circ$



⑧ i) Change the logarithm to a quotient of logarithms with the indicated base.

Base 6.

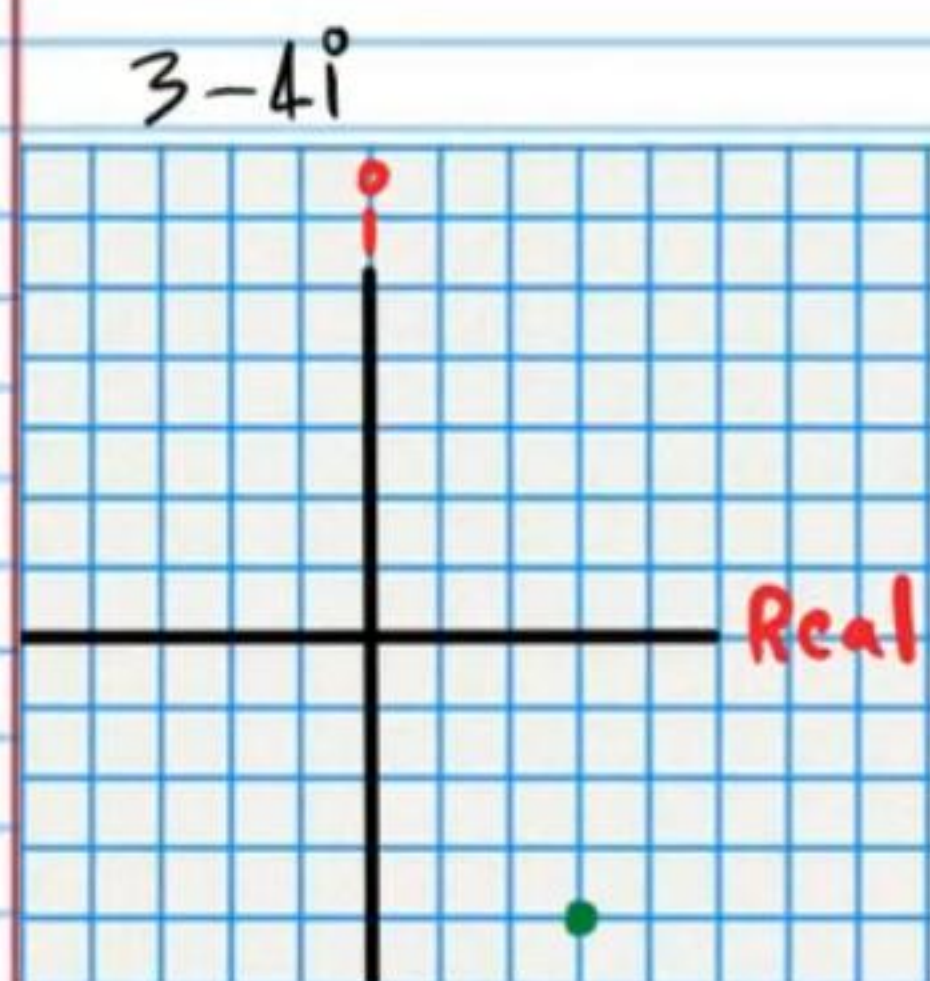
$\log_2 9 = \frac{\log_6 9}{\log_6 2}$

ii) Given the following intensity of sound, find the loudness perceived by the human ear.

10^{-3} W/m^2

$L(10^{-3}) = 10 \log (10^{12}(10^{-3})) \stackrel{I}{=} 10 \log_{10} (10^9) = 10(9) = 90 \text{ dB}$

iii) Graph the following complex number.



9i) Solve. Leave your answer as an exponential expression.

$$\frac{\log X}{2} - 10 = 1$$

$$\frac{\log_{10} X}{2} - 10 = 1$$

$$\frac{\log_{10} X}{2} = 11$$

$$\log_{10} X = 22$$

$$X = 10^{22}$$

iv) Solve. Round to the nearest thousandth.

$$3 = 10^{2x} \cdot 10^{5x}$$

$$3 \stackrel{I}{=} 10^{7x}$$

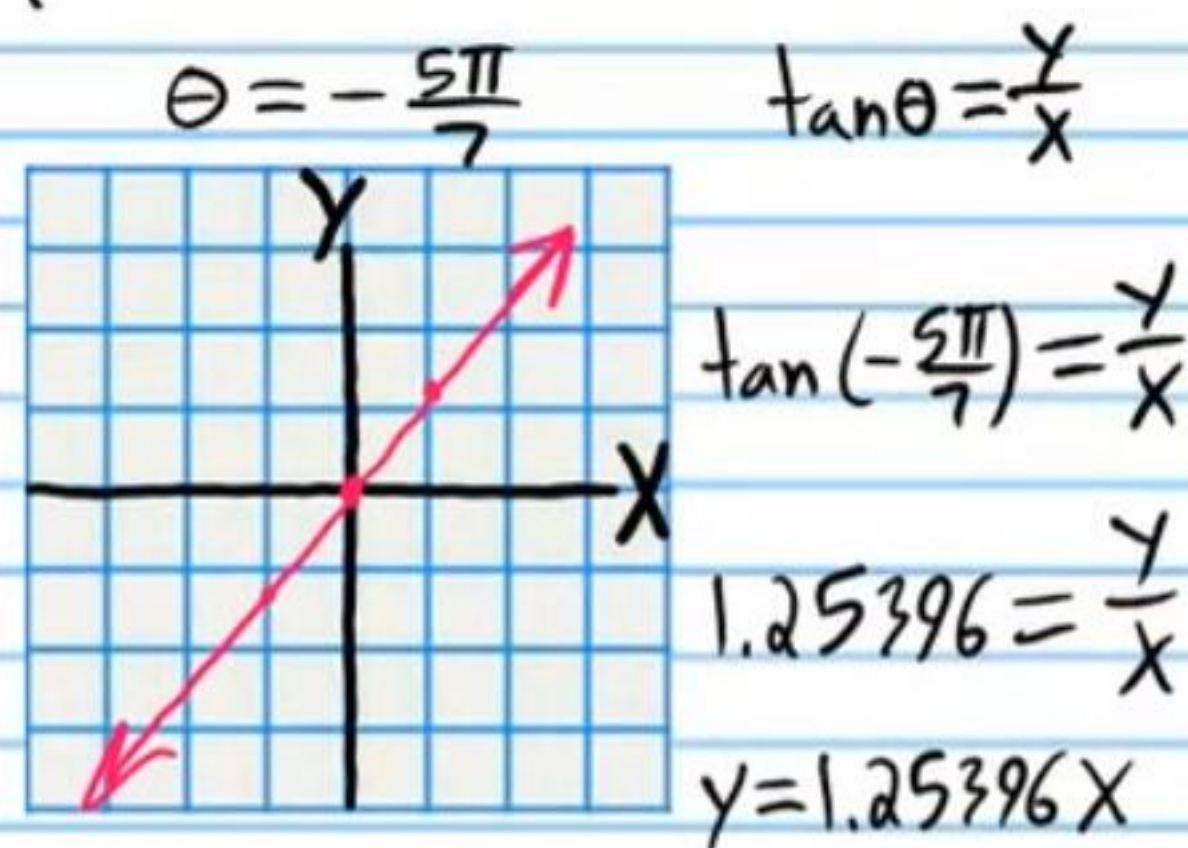
$$7x = \log_{10} 3$$

$$7x = \log 3$$

$$x = \frac{\log 3}{7}$$

$$x \approx 0.068$$

ii) Graph the following polar equation.



$$y = mx + b$$

10) Are the following functions inverses?

$$f(x) = \frac{x}{2} - 2 \quad \& \quad g(x) = 2(x-2)$$

$$f(g(x)) = f(2(x-2)) = \frac{2(x-2)}{2} - 2 = x-2-2 = x-4$$

No

11) Show algebraically that "f" and "f⁻¹" are inverse functions.

$$f(x) = 9(4^x) - 7 \quad \& \quad f^{-1}(x) = \log_4\left(\frac{x+7}{9}\right)$$

$$f(f^{-1}(x)) = f\left(\log_4\left(\frac{x+7}{9}\right)\right) = 9\left(4^{\log_4\left(\frac{x+7}{9}\right)}\right) - 7$$

$$= 9\left(\frac{x+7}{9}\right) - 7$$

$$= x+7-7$$

$$= x$$

$$f^{-1}(f(x)) = f^{-1}(9(4^x) - 7) = \log_4\left(\frac{9(4^x) - 7 + 7}{9}\right) = \log_4\left(\frac{9(4^x)}{9}\right)$$

$$= \log_4 4^x$$

$$= x$$

⑫ If $f(x) = x^2 + 12x - 27$ & $h(x) = 2x^2$, find the coordinates of the point(s) where the graphs of these functions intersect.

$$f(x) = h(x)$$

$$x^2 + 12x - 27 = 2x^2$$

$$0 = x^2 - 12x + 27$$

$$0 = (x-3)(x-9)$$

$$x=3 \text{ or } x=9$$

$$h(3) = 2(3)^2 = 18$$

$$\boxed{(3, 18)}$$

$$h(9) = 2(9)^2 = 162$$

$$\boxed{(9, 162)}$$

⑬ If a \$1,029 investment grows to \$28,000 in 25 years, what interest rate was applied if it was compounded continuously? Round the nearest hundredth OF A PERCENT.

$$A = Pe^{rt}$$

$$28,000 = 1,029e^{r(25)}$$

$$\frac{28,000}{1,029} = \frac{1,029e^{25r}}{1,029}$$

$$27.21088... = e^{25r}$$

$$25r = \log_e 27.21088...$$

$$25r = \ln 27.21088...$$

$$r = \frac{\ln 27.21088...}{25}$$

$$r = \frac{3.3036...}{25}$$

$$r = 0.132144...$$

$$\boxed{13.21\%}$$

⑭ If $f(x) = \frac{\sqrt{4x+1}}{2} - 8$, and $f(k) = 2$, find k .

$$f(x) = \frac{\sqrt{4x+1}}{2} - 8$$

$$f(k) = \frac{\sqrt{4k+1}}{2} - 8$$

$$2 = \frac{\sqrt{4k+1}}{2} - 8$$

$$10 = \frac{\sqrt{4k+1}}{2}$$

$$20 = \sqrt{4k+1}$$

$$20^2 = (\sqrt{4k+1})^2$$

$$400 = 4k+1$$

$$399 = 4k$$

$$\boxed{\frac{399}{4} = k}$$

⑮ If $f(x) = -\frac{6}{1-3x}$ and $f(n) = -1$, find n .

$$f(x) = -\frac{6}{1-3x}$$

$$f(n) = -\frac{6}{1-3n}$$

$$-1 = -\frac{6}{1-3n}$$

$$-(1-3n) = -6$$

$$-1+3n = -6$$

$$3n = -5$$

$$\boxed{n = -\frac{5}{3}}$$

Find an equation of the inverse of each function below.

⑩ $f(x) = -\frac{5^x}{3} + 4$

I) $y = -\frac{5^x}{3} + 4$

II) $x = -\frac{5^y}{3} + 4$

III) $x - 4 = -\frac{5^y}{3}$

$$-3(x - 4) = 5^y$$

$$-3x + 12 = 5^y$$

$$y = \log_5(-3x + 12)$$

IV) $f^{-1}(x) = \log_5(-3x + 12)$

⑪ $f(x) = 7 - 3x^3$

I) $y = 7 - 3x^3$

II) $x = 7 - 3y^3$

III) $x - 7 = -3y^3$

$$\frac{x - 7}{-3} = y^3$$

$$\sqrt[3]{\frac{x - 7}{-3}} = \sqrt[3]{y^3}$$

$$y = \sqrt[3]{\frac{x - 7}{-3}}$$

IV) $f^{-1}(x) = \sqrt[3]{\frac{x - 7}{-3}}$

⑫ If $f(x) = |x| - 2$, what is the average rate of change of the function per unit x from $x = -1$ to $x = 1$?

$$M_{ave} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$M_{ave} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$M_{ave} = \frac{f(1) - f(-1)}{1 - (-1)}$$

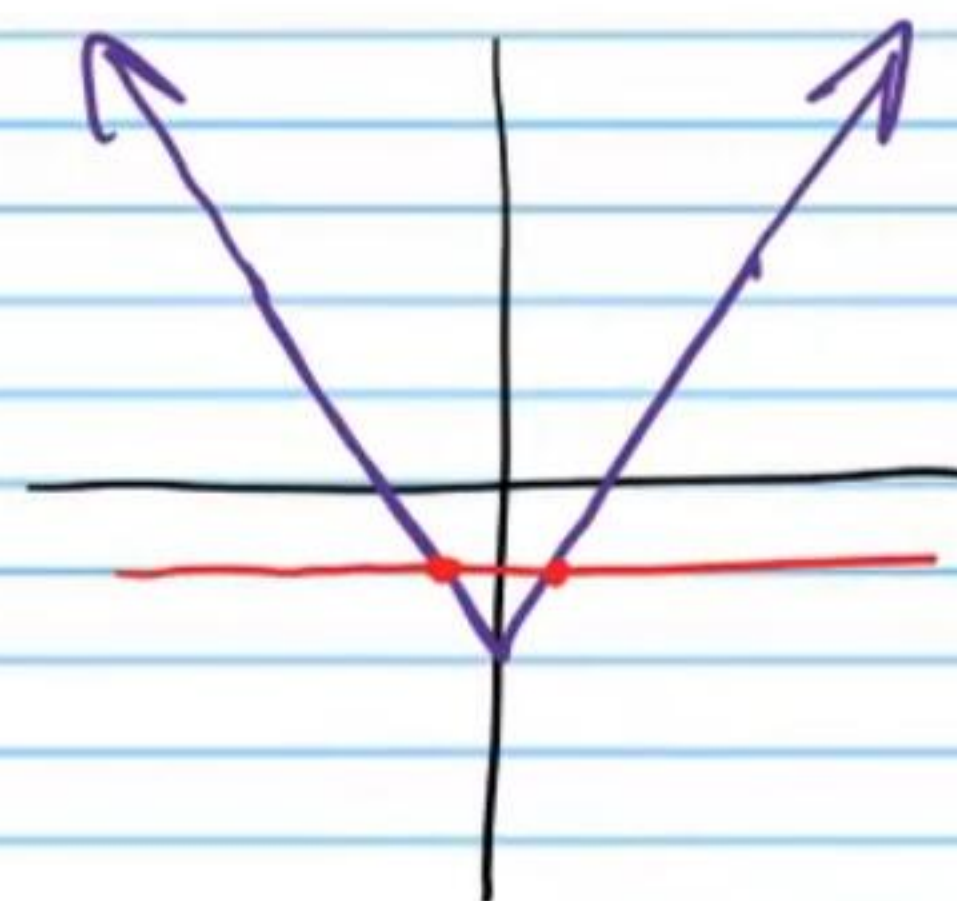
$$M_{ave} = \frac{|1| - 2 - (|-1| - 2)}{1 + 1}$$

$$M_{ave} = \frac{-1 - (-1)}{2}$$

$$M_{ave} = \frac{-1 + 1}{2}$$

$$M_{ave} = \frac{0}{2}$$

$$M_{ave} = 0$$



(19) If $f(x) = -3x^2 + x + 3$ and $f(P) = -3$, find P.

$$f(x) = -3x^2 + x + 3$$

$$f(P) = -3P^2 + P + 3$$

↓

$$-3 = -3P^2 + P + 3$$

$$0 = -3P^2 + P + 6$$

$$P = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$P = \frac{-1 \pm \sqrt{1^2 - 4(-3)(6)}}{2(-3)}$$

$$P = \frac{-1 \pm \sqrt{1 - (-72)}}{-6}$$

$$P = \frac{-1 \pm \sqrt{73}}{-6}$$

$$P = \frac{1 \pm \sqrt{73}}{6}$$

(20) Solve. Round to the nearest ten thousandth.

$$8 = 2 \ln x - 4$$

$$8 = 2 \log_e x - 4$$

$$12 = 2 \log_e x$$

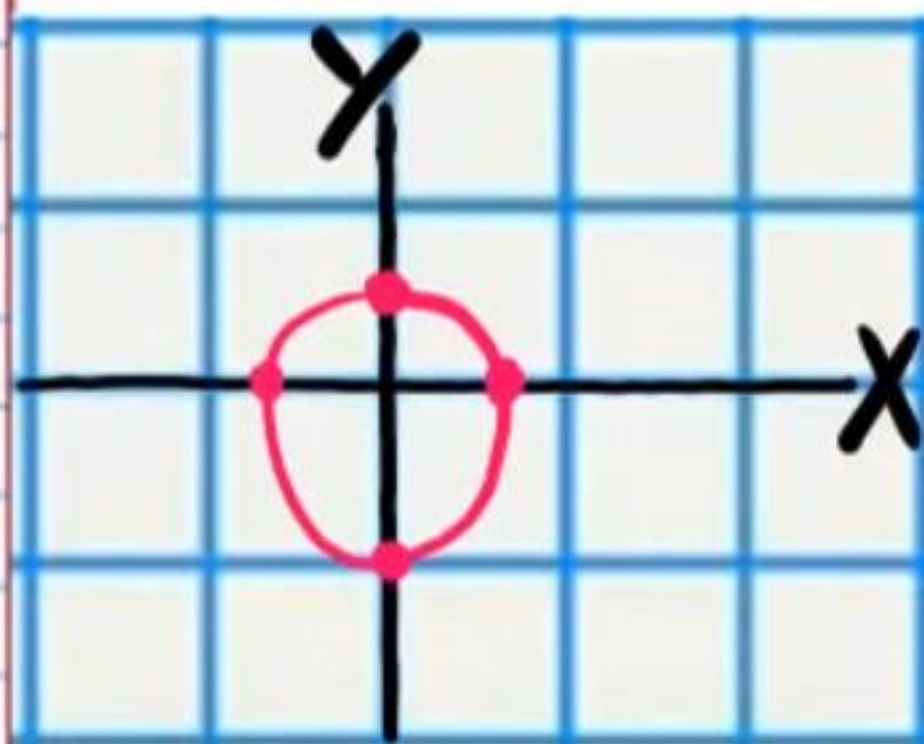
$$6 = \log_e x$$

$$e^6 = x$$

$$x \approx 403.429$$

(21) Graph the following equation using radian units. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

$$r = \frac{2}{3 + \sin \theta}$$



1st Point

$$\theta = 0$$

$$r = \frac{2}{3 + 0}$$

$$r = \frac{2}{3}$$

$$\left(\frac{2}{3}, 0\right)$$

2nd Point

$$\theta = \frac{\pi}{2}$$

$$r = \frac{2}{3 + 1}$$

$$r = \frac{2}{4}$$

$$r = \frac{1}{2}$$

$$\left(\frac{1}{2}, \frac{\pi}{2}\right)$$

3rd Point

$$\theta = -\frac{\pi}{2}$$

$$r = \frac{2}{3 - 1}$$

$$r = 1$$

$$\left(1, -\frac{\pi}{2}\right)$$

4th Point

$$\theta = \pi$$

$$r = \frac{2}{3 + 0}$$

$$r = \frac{2}{3}$$

$$\left(\frac{2}{3}, \pi\right)$$

22) The half life of lead-210 is 22.2 years. If a scientist has 0.45 grams, how much will be left after 100 years? Round to the nearest thousandth.

$$I) A(t) = A_0 e^{kt}$$

$$II) A(t) = A_0 e^{kt}$$

$$\frac{1}{2} A_0 = A_0 e^{k(22.2)}$$

$$A(100) = 0.45 e^{(-0.03122 \dots)(100)}$$

$$A(100) = 0.45 e^{-3.1222 \dots}$$

$$A(100) = 0.45 (0.04405 \dots)$$

$$A(100) \approx 0.02 \text{ grams}$$

$$\frac{\frac{1}{2} A_0}{A_0} = \frac{A_0 e^{22.2k}}{A_0}$$

$$\frac{1}{2} = e^{22.2k}$$

$$22.2k = \log_e\left(\frac{1}{2}\right)$$

$$22.2k = \ln\left(\frac{1}{2}\right)$$

$$k = \frac{\ln\left(\frac{1}{2}\right)}{22.2}$$

$$k = \frac{-0.693 \dots}{22.2}$$

$$k \approx -0.03122 \dots$$

23) The population of a colony of bacteria after "t" hours can be predicted with the equation below.

$$P(t) = \frac{156,213}{1 + 19e^{-0.2t}}$$

i) What is the carrying capacity of the population?

$$156,213 \text{ bacteria}$$

ii) What is the initial population?

$$P(0) = \frac{156,213}{1 + 19e^{-0.2(0)}} = \frac{156,213}{1 + 19e^0} \stackrel{III}{=} \frac{156,213}{1 + 19(1)}$$

$$= \frac{156,213}{1 + 19}$$

$$\approx 7,811 \text{ bacteria}$$

24) i) Using the equation in problem #23, what will the population of bacteria be after 5 hours?

$$P(5) = \frac{156,213}{1 + 19e^{-0.2(5)}} \approx 19,552 \text{ bacteria}$$

ii) Demonstrate how to find the value of the logarithm below if your calculator is programmed to compute base 10 or base "e" logarithms only. Round to the nearest thousandth.

$$\log_3 14 = \frac{\ln 14}{\ln 3} \approx 2.402$$

Q5) The population of a colony of bacteria follows the law of uninhibited growth. If the size of the colony quadruples in 5 days, how long will it take for 26 bacteria to increase to 1,500 bacteria?

I) $A(t) = A_0 e^{kt}$

$$4A_0 = A_0 e^{k(5)}$$

$$\frac{4A_0}{A_0} = \frac{A_0 e^{5k}}{A_0}$$

$$4 = e^{5k}$$

$$5k = \log_e 4$$

$$5k = \ln 4$$

$$k = \frac{\ln 4}{5}$$

$$k = 0.2772...$$

II) $A(t) = A_0 e^{kt}$

$$1500 = 26 e^{(0.2772...)t}$$

$$57.692... = e^{(0.2772...)t}$$

$$(0.2772...)t = \log_e(57.692...)$$

$$(0.2772...)t = \ln(57.692...)$$

$$(0.2772...)t = 4.0551...$$

$$t \approx 14.626 \text{ days}$$